

Problem 3

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Problem 1 Suppose that $\int_1^3 f(x) dx = 4$

(a) Evaluate the following integrals.

(i) $\int_0^2 f(x+1) dx$

(ii) $\int_1^9 3 \frac{f(\sqrt{x})}{\sqrt{x}} dx$

(iii) $\int_{\frac{1}{3}}^1 f(3x) dx$

(iv) $\int_2^4 3f(x-1) dx$

(v) $\int_0^{\sqrt{2}} 3xf(x^2+1) dx$

(b) Assume that f is odd. Evaluate $\int_{-3}^{-1} f(x) dx$

(c) Assume that f is even. Evaluate $\int_{-3}^{-1} f(x) dx$